Motorcycle Lane Filtering and Lane Splitting in the United States: Protocol, Search Results, and Preliminary Findings

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Abstract Lane filtering, in which a motorcyclist advances between lanes of stopped or very slow traffic, has been expressly authorized by five U.S. states since 2019, yet the evidence on its safety effects is scattered and its legal status is often misreported. This paper reports the protocol, search results, and preliminary findings of a PRISMA 2020 systematic review of motorcycle lane filtering and lane splitting, with a completed primary-source census of the law in all 50 states and the District of Columbia. Scopus, Web of Science, and TRID were searched on 6 October 2026 with a strategy tested against 13 known records and revised once, yielding 178 records (114 after duplicate removal). Screening, agency-report retrieval, and data extraction are in progress. A preliminary synthesis of 12 seed studies indicates that the speed differential relative to surrounding traffic is the clearest modifier of injury severity, that the principal filtering conflict is the same-direction sideswipe, that familiarity with filtering does not ensure knowledge of the law, and that three government evaluations of legalization report mixed or null crash effects. The census finds five states that expressly authorize filtering under different conditions, one that defines lane splitting without authorizing it, 30 that expressly prohibit both maneuvers, 13 with no motorcycle-specific provision, and two with agency guidance only, and documents errors in secondary summaries for nine jurisdictions. Four gaps are identified: no longitudinal evaluation of a U.S. law change, no exposure data, little evidence linking legal knowledge to behavior, and no comparison across statutes.

Keywords: Motorcycle Lane Filtering; Lane Splitting; Motorcycle Safety; Systematic Review; PRISMA; Traffic Safety Legislation

1 Introduction

Motorcyclists accounted for 6,335 of the 40,901 people killed on U.S. roads in 2023, about 15 percent of all traffic deaths, and died at roughly 28 times the rate of passenger-car occupants per vehicle mile traveled [28]. Nearly two-thirds of those deaths occurred in urban areas, where motorcycles share congested roadways with larger vehicles. In that setting, a rider stopped in a queue is exposed to a particular hazard: being struck from behind by a vehicle whose driver has not registered the motorcycle. Lane filtering, in which a motorcycle moves slowly between or alongside stopped vehicles to the front of the queue, is advocated as a way to remove the rider from that position [22]. The same maneuver places the motorcycle beside vehicles whose drivers may not expect it, which is the mechanism behind the sideswipe conflicts that critics emphasize. Which of these effects dominates, and under what conditions, is an empirical question that U.S. policy has now made urgent.

Terminology matters, because the recent statutes regulate two behaviors differently (Figure 1). Lane filtering refers to passing traffic that is stopped or moving very slowly; lane splitting refers to passing between lanes of moving traffic. Utah, Montana, Arizona, Colorado, and Minnesota have enacted statutes since 2019 that expressly authorize filtering under bounded conditions, typically a speed cap on the motorcycle and a requirement that the overtaken traffic be stopped or slow, while Minnesota also authorizes low-speed splitting [5, 12, 24, 26, 40]. California, where the practice has long been lawful, defined lane splitting in statute in 2016 without a speed limit [10]. Two of the five new statutes carry sunset clauses that require legislative renewal in 2027, and the AAA Foundation for Traffic Safety has identified the evaluation of these laws as a research need [1], so the question of whether they improve safety is before policymakers now.

The evidence base is small and uneven. The only large crash-based study known to the authors comes from California and concerns lane splitting at freeway speeds [35]; naturalistic and trajectory studies come from France, Australia, Thailand, and India, in traffic environments that differ from U.S. conditions [7, 27, 32, 36]; survey evidence on awareness and attitudes comes mainly from Australia [8, 38]; and the before-and-after evaluations of legalization known to the authors before the search are government reports from New South Wales, Western Australia, and France [11, 38, 42]. Earlier syntheses were narrative. A 2010 literature review for the Oregon Department of Transportation found the research base nearly non-existent [37]; a 2017 agency summary for the Washington Traffic Safety Commission reviewed the same handful of studies [43]; and a 2016 review by an advocacy organization relied heavily on the California findings [22]. To our knowledge, no systematic review of lane filtering or lane splitting has been published, and no primary-source census of U.S. state law exists; the state-by-state summaries in circulation are produced by insurers, law firms, and advocacy

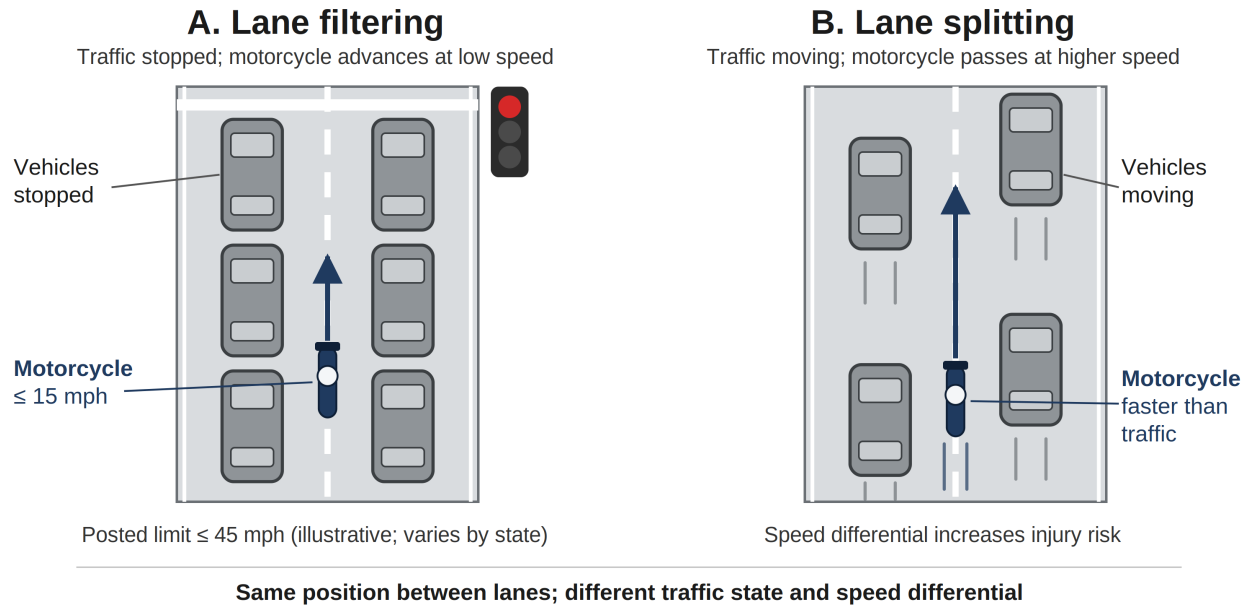


Figure 1: Lane filtering (A) versus lane splitting (B): the same position between lanes under different traffic states and speed differentials (illustrative conditions; statutory thresholds vary by state).

groups and, as the census reported here shows, do not always agree with one another or with the statutes.

This paper therefore has two aims. The first is a systematic review, following PRISMA 2020 [30], of the empirical evidence on the safety, behavioral, and attitudinal effects of motorcycle lane filtering and lane splitting, including government evaluations of legalization. The second is a census of the statutory status of both maneuvers in all 50 states and the District of Columbia, coded from the primary legal text with every discrepancy against secondary sources logged; the census is complete and is reported in full in this version. The review is organized around four research questions on crash and injury outcomes, rider and driver behavior, road-user awareness and attitudes, and the effects of legalization; the census answers a fifth question on current legal status.

This version of the paper is a stage-one report. It presents the complete protocol, the database search and its validation, the deduplicated record set that will enter screening, a preliminary synthesis of the studies that were known before the systematic search and that motivated it, and the completed 51-jurisdiction legal census. It is posted before title and abstract screening begins so that the protocol is on public record with a date. Screening, full-text assessment, data extraction, and the full synthesis will be reported in an updated version. Sections that depend on those steps are identified as such, and the findings reported here are labeled as preliminary wherever they rest on the seed set rather than on the systematic search.

2 Methodology

This paper combines a systematic literature review with a census of state law. The review was structured according to the PRISMA 2020 guidelines [30], including the PRISMA-S extension for reporting literature searches [34], and follows three stages: (1) identification, (2) screening and eligibility, and (3) data extraction and synthesis, the structure used in a recent systematic review in this journal [25]. Stage 1 is complete and is reported in full; stages 2 and 3 are described as protocol. The legal census followed a separate protocol, described at the end of this section. The review protocol was not registered on a review registry; this preprint, posted before screening began, serves as the dated public record of the protocol.

2.1 Research Questions

Four questions are addressed through a systematic search of the empirical literature and a fifth through a separate legal census. RQ1: What crash, injury, and conflict outcomes have been reported for motorcycle lane filtering and lane splitting? RQ2: What rider and driver behaviors have been observed during these maneuvers? RQ3: What is known about road users' awareness of the law, legal knowledge, and attitudes toward the practice? RQ4: What crash, behavioral, and implementation effects have been reported in jurisdictions that legalized or trialed the maneuver? RQ5: What is the current statutory status of lane filtering and lane splitting in each U.S. state and the District of Columbia? Lane filtering (passing stopped or very slow traffic) and lane

splitting (passing moving traffic) were treated as distinct behaviors throughout, because recent U.S. statutes regulate them differently [24, 39].

2.2 Search Strategy

Scopus, Web of Science Core Collection, and the Transportation Research International Documentation database (TRID) were searched. In Web of Science, all ten editions of the Core Collection available through the university subscription were selected (SCI-EXPANDED, SSCI, AHCI, CPCI-S, CPCI-SSH, BKCI-S, BKCI-SSH, ESCI, CCR-EXPANDED, and IC). The first two are the largest multidisciplinary citation indexes and were used by comparable reviews in this journal [15, 33]; TRID was added because much of the relevant evidence consists of agency reports and conference papers that are not indexed elsewhere. No start-year limit was applied, since the literature is small and early work remains relevant; records were limited to English. No document-type filter was applied at the database stage so that every exclusion could be reported at screening.

Table 1 lists the search strings, dates, and number of records retrieved from each database for both versions of the strategy. Each string combined a block of maneuver terms with a block of vehicle terms; the word "filtering" was never used without a motorcycle term, because doing so retrieves signal-processing studies. Because TRID supports limited Boolean logic, its searches were run separately and are listed individually. The first version (v0.1) was run on 1 and 2 October 2026 by one author (TIC), who inspected every retrieved record and classified it as potentially relevant or as one of four kinds of noise (bicycle, car or automated-vehicle lane sharing, signal or data filtering, other).

Before full screening, the strategy was tested against 13 database records of nine studies known to be relevant, each searched by exact title in the database where it was expected (the validation set is reported in the Results). All five TRID records and six of the eight Scopus and Web of Science records were retrieved by v0.1; the two misses were the Scopus and Web of Science records of one study [36], which was indexed but whose title and abstract refer to "filtering maneuvers" without the word "lane". Inspection of the retrieved titles also showed that "seepage" is used for the same behavior in South Asian traffic studies [3]. The strategy was therefore revised (v0.2) to add filtering-maneuver terms, "seepage", and a proximity clause, and re-run on 6 October 2026. Version 0.2 retrieved all 13 validation records and is the strategy reported in Figure 2 and used for all counts in this paper; the complete strings for both versions are given in Appendix A. One further relevant article, identified during validation, was found not to be indexed in Scopus or Web of Science at all and will be captured through citation searching.

Table 1: Search Strategy by Database and Version

Database	Version	Search string (abridged; full strings in Appendix A)	Date run	Records
Scopus	v0.1	TITLE-ABS-KEY((maneuver block: "lane filtering" OR "lane splitting" OR "lane sharing" OR "filtering through traffic" OR "inter-file" OR "white lining", with hyphenated variants) AND (vehicle block: motorcycl* OR motorbike* OR "powered two-wheeler*" OR PTW* OR "two-wheeler*" OR moped* OR scooter*))	1 Oct 2026	19
Web of Science Core Collection	v0.1	TS=(same maneuver block AND same vehicle block)	1 Oct 2026	22
TRID	v0.1	Four searches: "lane splitting"; "lane filtering"; "lane sharing" motorcycle; filtering motorcycle	2 Oct 2026	22; 6; 4; 34
Scopus	v0.2	v0.1 maneuver block OR "filtering maneuver*" OR "filtering manoeuvre*" OR "filtering behavio*" OR seepage OR (filtering W/3 (motorcycl* OR "two-wheeler*" OR PTW*)), AND the vehicle block	6 Oct 2026	41
Web of Science Core Collection	v0.2	Same additions as Scopus, with NEAR/3 in place of W/3	6 Oct 2026	68
TRID	v0.2	The four v0.1 searches re-run, plus: seepage motorcycle OR seepage "two-wheeler"	6 Oct 2026	22; 6; 4; 36; 1

Note: All searches limited to English; no start-year or document-type limit. Records counted before duplicate removal. TRID searches were run on all fields with Language = English and were sorted by date because TRID offers no relevance sort. Searcher: TIC.

2.3 Other Sources

Government and agency evaluations are sought separately, because the before-and-after evaluations of legalization in the seed set are agency publications, and because none of the three agency evaluations known to the authors before the search [11, 38, 42] was retrieved by any of the seven searches or appears in the record set. The list of sources was fixed before searching so that the grey-literature track would not be selective. It comprises the websites of NHTSA; the California Highway Patrol; the Utah Department of Public Safety;

the Arizona Department of Transportation and Department of Public Safety; the Colorado Department of Transportation; the Montana Department of Justice, Motor Vehicle Division; the Minnesota Department of Public Safety, Office of Traffic Safety; Transport for NSW; the Queensland Department of Transport and Main Roads; the Western Australia Road Safety Commission; Cerema and ONISR (France); and SafeTREC at the University of California, Berkeley. Three professional and advocacy organizations (American Motorcyclist Association, Motorcycle Safety Foundation, Ride to Work) will be searched and recorded separately because their evidentiary standing differs from that of government sources. The reference lists of three prior reviews [22, 37, 43] and of all included studies will be screened, and the first 100 Google Scholar results by relevance for "lane filtering" and "lane splitting" will be screened as a supplementary source, following the practice of recent reviews in this journal [2, 25]. These records enter the flow diagram through the "identification via other methods" column (Figure 2). None of these searches had been run at the time of writing.

2.4 Eligibility Criteria

Studies are eligible if they are empirical, whether crash-based, naturalistic, observational, trajectory or surrogate-safety, survey, experimental, simulation, or policy-evaluation designs, and examine motorcycle or powered-two-wheeler riding between lanes or past same-direction traffic as a studied behavior or analyzed subgroup, reporting safety, crash, injury, conflict, rider or driver behavior, awareness or attitudinal, or legalization outcomes. The evaluation design of each study is recorded as an extraction field. Empirical government reports are eligible. Narrative and systematic reviews are excluded from synthesis but used for citation searching. Records are excluded for one of seven coded reasons: E1, not about motorcycles or powered two-wheelers (for example, bicycles, automated-vehicle lane sharing, or signal-processing "filtering"); E2, motorcycles but not riding between lanes; E3, not empirical; E4, operational or capacity-only outcome with no safety, crash, or behavioral measure; E5, not in English; E6, abstract unavailable; E7, duplicate or superseded version of an included study.

2.5 Study Selection

Records from the seven v0.2 searches were compiled in a single workbook by one author (IJ) and deduplicated by matching digital object identifiers and then exact titles, with the first occurrence retained in the order Scopus, Web of Science, TRID; one pair of records with identical titles but different venues, years, and page ranges was checked by hand and kept as two records. Duplicate removal is reported in Figure 2. Two reviewers will independently screen a calibration set of 50 titles and abstracts, compare decisions, and agree on the application of each criterion before screening the remaining records independently. Disagreements and borderline cases will be discussed by the full author team until consensus is reached, the deliberative approach used in a recent review in this journal [25]; percent agreement on the calibration set will be reported, but no chance-corrected statistic will be computed, because the small number of records makes such a statistic unstable and the consensus procedure resolves every disagreement. Full texts of all records retained at this stage will be assessed against the same criteria by the same two reviewers, and reasons for exclusion will be coded; counts by reason will be reported in the updated flow diagram.

2.6 Data Extraction and Quality Appraisal

For each included study, one reviewer will extract and a second will verify: country and traffic setting; study design and evaluation design; sample; the maneuver studied (filtering, splitting, or mixed); outcome measures; principal findings; and limitations, using a structured form whose fields are fixed in the project workbook. No formal quality score will be assigned, the practice followed by most systematic reviews in this journal [21, 25]; instead the design, sample, and evidentiary status (peer-reviewed, government, or advocacy) of each study are reported in the extraction table and taken into account in the synthesis.

2.7 Legal Census

For each of the 50 states and the District of Columbia, one author (SS, with one worked example coded by IJ) read the governing statute or regulation on the official state legislature or code website, recorded a second official source (state DOT, DPS, or licensing-agency manual or FAQ) where one existed, and noted the access date. Where the official site could not be reached with the tools available to the coder, the row was marked pending until a second author (IJ) read the section on the official site; this was needed for 13 jurisdictions and in every case the official text matched the text that had been read on a mirror site. A 50-state table compiled before the census served as a starting dataset, not as ground truth. Unlike earlier state-practice reviews in this journal that relied on surveys of agency staff, the census is based on the primary legal text, which is verifiable, dated, and free of respondent error. Lane filtering and lane splitting were coded separately into one of five categories: (A) expressly authorized in primary law or regulation, with conditions recorded in

separate fields (motorcycle speed cap, condition of the overtaken traffic, posted-limit cap, minimum number of lanes, geometry, excluded locations, and effective and sunset dates); (B) defined or recognized in primary law without an express authorization provision; (C) expressly prohibited in primary law or regulation; (D) no specific provision identified, general lane-use rules only; and (E) official agency guidance without a specific governing provision. Category D is not a prohibition: it records the absence of a motorcycle-specific provision. Category C includes provisions that bar passing within the same lane without naming riding between lanes (Massachusetts, Virginia); these are identified in the table. Where a statute itself names the maneuver, the statute's terminology governed the coding. Every discrepancy between the starting dataset, secondary sources, and the primary text was recorded in a conflict log and resolved against the primary text. A ten-jurisdiction pilot was carried out first to test the coding scheme, and the remaining jurisdictions were coded with the same scheme between 2 and 7 October 2026; the Montana worked example was first read on 26 September 2026. Effective dates were recorded only where the official history line states a date; where it gives only a session law or year, the date field was left blank rather than estimated. The census is a complete enumeration of all 51 jurisdictions rather than a sample, unlike case-based legal analyses in this journal [17]. It is limited to statutory and regulatory text: case law, local ordinances, and enforcement practice are not reviewed, and the coding describes what the text provides rather than offering legal advice.

2.8 Synthesis

Because the eligible studies differ in design, setting, and outcome measure, and because several come from mixed-traffic environments without a U.S. counterpart, results will first be described as a set (year, country, traffic setting, design, and maneuver studied), following the descriptive analysis used in recent reviews in this journal [21, 25], and then synthesized thematically by research question. Filtering and splitting evidence will be kept distinct, and government evaluations will be presented alongside peer-reviewed studies with their evidentiary status identified. The legal census is reported as a table and a two-panel map (filtering and splitting), with the conflict log summarized in the text. No meta-analysis is planned. In this version, the thematic synthesis is applied to the seed set of studies that were known before the systematic search; it is labeled as preliminary and will be replaced by the synthesis of the included studies.

3 Results

3.1 Search Results and Record Set

The seven v0.2 searches returned 178 records: 41 from Scopus, 68 from Web of Science, and 69 from the five TRID searches (Figure 2; Table 2). Duplicate matching removed 64 records, leaving 114 unique records for title and abstract screening. Of the 114, 36 were found in more than one database (22 in Scopus and Web of Science, 12 in all three, and 2 in Web of Science and TRID), while 39 were found only in TRID, 32 only in Web of Science, and 7 only in Scopus. The 39 TRID-only records include the two California studies [29, 35], the two prior U.S. reviews [22, 37], and several Australian and French conference papers, which confirms the value of a transportation-specific index for this topic.

Table 2: Records Retrieved by Each v0.2 Search and the Searcher's Pre-Screening Inspection

Search	Database	Query	Records	Potentially relevant	Noise: bicycle	Noise: car or AV	Noise: signal filtering	Noise: other	Unique records contributed
SL-007	Scopus	v0.2 string	41	26	0	0	4	11	41
SL-008	Web of Science	v0.2 string	68	39	1	1	10	17	34
SL-009	TRID	"lane splitting"	22	20	0	0	1	1	14
SL-010	TRID	"lane filtering"	6	6	0	0	0	0	2
SL-011	TRID	"lane sharing" motorcycle	4	3	0	0	0	1	0
SL-012	TRID	filtering motorcycle	36	21	0	1	10	4	22
SL-013	TRID	seepage motorcycle OR seepage "two-wheeler"	1	1	0	0	0	0	1
Total			**178**	**116**	**1**	**2**	**25**	**34**	**114**

Note: Every retrieved record was inspected by the searcher (TIC) and classified before screening; "potentially relevant" is that inspection, not a screening decision, and counts overlap across searches. "Unique records contributed" is the number of records first retrieved by that search when duplicates are removed in the order listed; the column sums to the 114 unique records.

The record set spans 1972 to 2026, with 75 of the 114 records published in 2015 or later and 44 in 2020 or later; 63 are journal articles, 40 conference papers, 6 reports, and 5 book chapters or other document types. The inspection of every retrieved record by the searcher, recorded before screening, gives an early

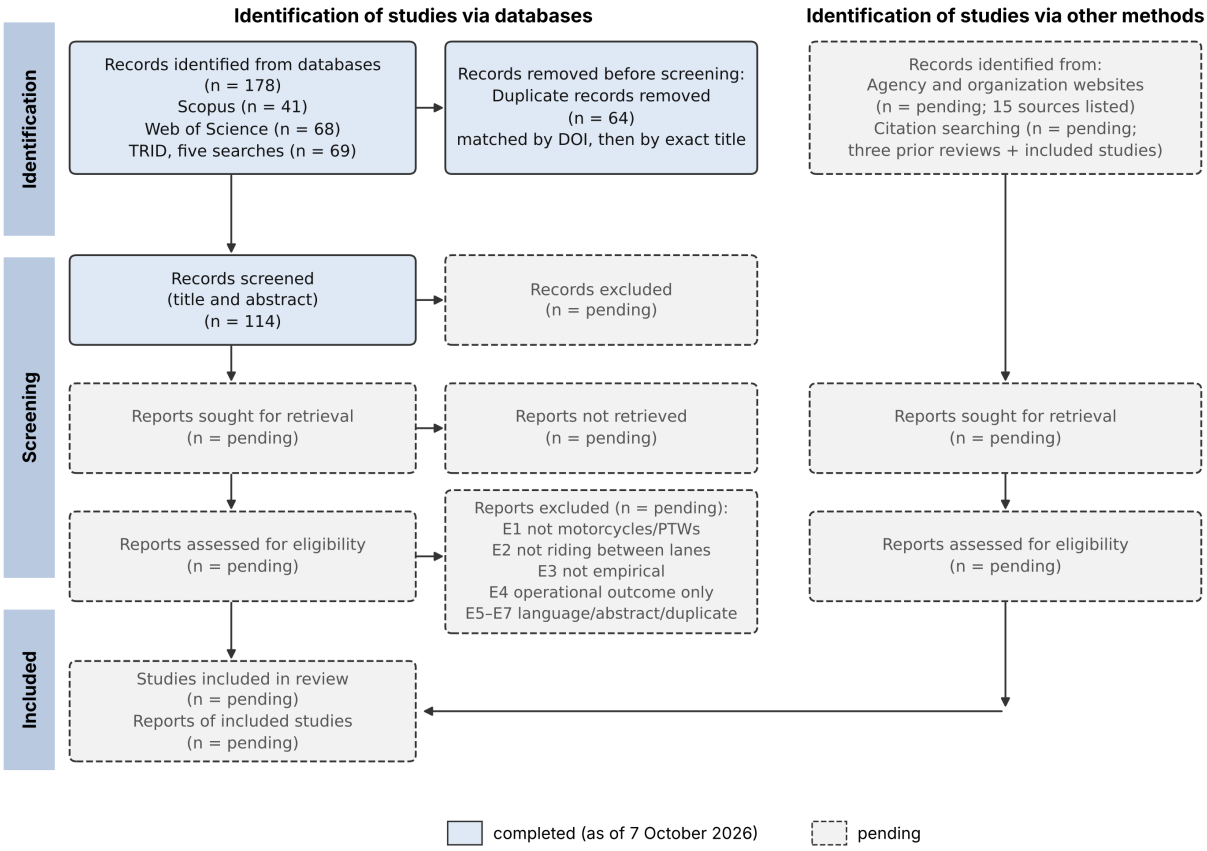


Figure 2: PRISMA 2020 flow diagram at the end of stage 1 (identification). Shaded boxes are complete as of 7 October 2026 (searches run 6 October, duplicates removed 7 October); dashed boxes are pending and will be filled in the updated version.

indication of what screening will find. In the v0.2 Scopus and Web of Science results, 26 of 41 and 39 of 68 records were judged potentially relevant, and the rest were noise introduced mainly by the proximity clause (exhaust-particle filters, Kalman and washout filters) and by the truncation "PTW*", which matches a brand of radiation dosimeter. In the TRID searches, the single-term search "filtering motorcycle" was the noisiest (21 of 36 potentially relevant) but it was also the only search that retrieved a trajectory study of filtering and overtaking on Athens arterials [41]. The TRID searches together retrieved several other relevant European studies that were not in the seed set, among them an analysis of filtering risk on urban motorways in France [18] and a before-and-after comparison of motorcycle crashes around the 2011 legalization of filtering in Belgium [23]. The last of these is a fourth government-sponsored evaluation of legalization that the preliminary synthesis below does not yet include. These noise categories correspond to exclusion code E1 and will be removed at screening rather than by further narrowing the search, because narrowing was judged more likely to lose relevant records than to save screening effort.

3.2 Validation of the Search Strategy

Table 3 reports the validation set. It deliberately included a report and three conference papers that are indexed only in TRID [8, 14, 29, 35], so that the transportation-specific index was tested as well as the two multidisciplinary ones. The two records missed by v0.1 are both the article by Samalla and Chunchu [36], the most direct quantitative evidence on conflict risk during filtering, and the miss was caused by terminology rather than by indexing. This is the reason the strategy was revised before screening rather than after, and it is the reason the search terms in Table 1 include forms of "filtering" that are not preceded by "lane". The validation also identified one article on filtering and crash exposure that is not indexed in Scopus or Web of Science at all; it will be captured through citation searching, and its absence from the indexes is itself a reminder that a database search alone cannot be exhaustive in this field.

Table 3: Validation of the Search Strategy Against 13 Known Records

Study	Year	Document type	Database checked	Retrieved by v0.1	Retrieved by v0.2	Note
Aupetit, Espié and Bouaziz	2015	Journal article	Scopus	Yes	Yes	
Papakostopoulos, Nathanael and Psarakis	2020	Journal article	Scopus	Yes	Yes	
Promraksa et al.	2022	Journal article	Scopus	Yes	Yes	
Samalla and Chunchu	2024	Journal article	Scopus	No	Yes	Indexed; title and abstract use “filtering maneuvers” without “lane”
Aupetit, Espié and Bouaziz	2015	Journal article	Web of Science	Yes	Yes	
Papakostopoulos, Nathanael and Psarakis	2020	Journal article	Web of Science	Yes	Yes	
Promraksa et al.	2022	Journal article	Web of Science	Yes	Yes	
Samalla and Chunchu	2024	Journal article	Web of Science	No	Yes	As above
Rice, Troszak and Erhardt	2015	Report	TRID	Yes	Yes	“lane splitting” search
Ouellet	2012	Conference paper	TRID	Yes	Yes	“lane splitting” search
Beanland et al.	2015	Conference paper	TRID	Yes	Yes	“lane filtering” and “filtering motorcycle” searches
Eyssartier, Lanfranchi and Ragot-Court	2018	Conference paper	TRID	Yes	Yes	“lane splitting” search
Bec-Gerion, Eyssartier and Granié	2025	Book chapter	TRID	Yes	Yes	“lane splitting” search

Note: Each record was searched by exact title in the database named. A further article (Roaming the Virtual Lanes: Motorcyclists’ Lane Filtering-Splitting Manoeuvres and Exposure to Road Crashes, 2020) was found to be indexed in neither Scopus nor Web of Science and is therefore not a test of the strategy; it will be captured through citation searching.

3.3 Preliminary Synthesis of the Seed Studies

The synthesis in this section covers the studies that were known to the authors before the systematic search and that motivated the review (Table 4). It is not the result of the systematic search; it will be replaced by the synthesis of the included studies once screening and extraction are complete. Of the 12 empirical studies and evaluations in Table 4, nine were retrieved by the v0.2 database search; the three government evaluations were not, and will enter through the other-methods track.

Table 4: Seed Studies Used for the Preliminary Synthesis

Study	Setting and data	Method	Key finding	Principal limitation	In v0.2 record set
Rice et al. [35]	California; 5,969 CHP-investigated motorcycle crashes, June 2012 to August 2013	Descriptive crash analysis; supplemental lane-splitting field	Splitting riders (17%) had lower head, torso, and fatal injury rates and were less often rear-ended; injury rose with speed differential above 15 mph	Case-only; no exposure; rider self-selection; high-speed splitting, not stopped-traffic filtering	Yes (TRID)
Ouellet [29]	California freeways; video observation with speed data; 900 in-depth 1976 to 1977 Los Angeles crash investigations	Observed prevalence versus crash-involved share	Splitting frequency falls sharply above 40 mph; under 1% of all motorcycle crashes and 7% of freeway crashes; 63% observed versus 29% of crashes	Dated crash data; small absolute numbers; few camera sites	Yes (TRID)
Aupetit et al. [7]	Paris region; 11 commuters, one month	Instrumented motorcycles with cameras; self-confrontation interviews	Filtering is routine and perceptually demanding; riders search for hazards and manage their own conspicuity	Small sample; descriptive; no safety outcome	Yes (all three)
Mulvihill et al. [27]	Melbourne; 25 experienced riders on an urban test route	On-road instrumented ride with concurrent verbal protocols	Filtering in 27% of 43 eligible queue opportunities; filtering riders attended more to signal timing and own actions, less to through and following traffic	Proof-of-concept; single route; few filtering opportunities	Yes (TRID)

Continued on next page

Table 4 (continued)

Study	Setting and data	Method	Key finding	Principal limitation	In v0.2 record set
Beanland et al. [8]	Australian Capital Territory; 249 drivers surveyed before a filtering trial	Structured questionnaire; logistic regression	61% saw filtering at least weekly; 28% wrongly believed it was legal; 70% said it should be illegal; support higher among riders and those who believed it legal	Cross-sectional; convenience sample; non-U.S. regulatory context	Yes (TRID)
Papakostopoulos et al. [31]	Greece; experienced riders, motorcycle-mounted eye tracking	Eye tracking, video, and retrospective commentary	Riders use surrounding vehicles to anticipate driver behavior as well as to hold position; riding between lanes carries a specific perceptual load	Small sample; lane splitting rather than low-speed filtering	Yes (all three)
Promraksa et al. [32]	Thailand; 30 riders at one signalized intersection	Onboard lateral-clearance sensor; multilevel regression	Speed, filtering side, adjacent-vehicle condition and type, and riding frequency predict accepted lateral clearance	Mixed traffic in a developing-country setting; single site	Yes (all three)
Samalla and Chunchu [36]	India; urban vehicle-trajectory data	Automated filtering detection; surrogate safety measure	Sideswipe conflicts dominate during filtering; sideswipe risk about three times non-filtering, mainly with large vehicles; no gap-speed relationship	Weak lane discipline; not transferable to U.S. conditions	Yes (Scopus, Web of Science); missed by v0.1
Bec-Gerion et al. [9]	France; second national experiment; rider survey	Social-acceptability survey	Acceptability varies with prior experience of the maneuver and reported compliance	Self-report; one country	Yes (all three)
Transport for NSW [38]	Sydney CBD; eight-week trial, March to April 2013; five treatment sites and one control site	Before-and-after video observation, traffic sensors, intercept and online surveys, police monitoring	63% of online respondents said they always or mostly filter versus about 20 to 30% observed; no crashes during the trial; 48% of riders felt safer, about a quarter less safe; pedestrians judged at greatest risk	Eight-week period; small trial area; no exposure adjustment; crash outcome not estimable	No (agency report)
WARSC [42]	Western Australia; review 18 or more months after March 2021 legalization	Crash and offence data; community survey with knowledge test; stakeholder consultation	No notable change in rear-end or sideswipe motorcycle crashes; large share of community lacked accurate knowledge of the law; almost half of riders changed behavior to comply	Limited crash data; short post-period; descriptive	No (agency report)
Cerema [11]	France; 11 departments, February 2016 to January 2021; motorways and dual carriageways posted 70 km/h or more	Before-and-after crash comparison with control zone; about 4,500 police reports; behavior and acceptability studies	Two-wheeler crashes +12% on the experimental networks versus -10% across the experimental departments overall; compliance with the 50 km/h cap improved from 23% to 40%; results described as mixed	Small crash counts; behavior and exposure changed with publicity; not conclusive enough to codify	No (agency report)

Note: The seed set comprises the empirical studies and agency evaluations known to the authors before the systematic search. The last column reports whether the study appears among the 114 unique records retrieved by the v0.2 database search.

RQ1: Crash, injury, and conflict outcomes. The center of gravity of the crash-based literature in the seed set is a single descriptive study of California crash data. Rice et al. [35] analyzed 5,969 motorcycle-involved collisions investigated by the California Highway Patrol between June 2012 and August 2013, using a supplemental data element that recorded whether the rider was lane splitting at the time of the collision. Of these, 997 (17 percent) were lane splitting. Compared with other crash-involved riders, lane-splitting riders were less likely to sustain head injury (9 versus 17 percent), torso injury (19 versus 29 percent), or fatal injury (1.2 versus 3.0 percent), and were less likely to have been rear-ended (2.6 versus 4.6 percent). They also differed systematically as a population: they were more often riding on weekdays during commute hours, more likely to wear full-face helmets (81 versus 67 percent), traveling at lower speeds, less likely to have used alcohol, and less likely to carry a passenger. Injury likelihood rose with the speed differential between the

motorcycle and surrounding traffic; about 69 percent of splitters exceeded surrounding traffic by 15 mph or less, but 14 percent exceeded it by 25 mph or more, and injury outcomes worsened markedly in that group. Earlier work by Ouellet [29] combined freeway video observation with speed data and a re-analysis of 900 in-depth Los Angeles crash investigations from 1976 to 1977. He reported that the frequency of lane splitting declines sharply once average traffic speed exceeds about 40 mph, that lane-splitting crashes made up less than 1 percent of all motorcycle crashes and about 7 percent of freeway crashes in those data, and that splitting motorcycles were under-represented among crashes relative to their observed share of heavily congested freeway traffic (63 percent observed versus 29 percent of crashes). Outside the United States, the trajectory study by Samalla and Chunchu [36] in Indian mixed traffic found that sideswipe conflicts, particularly with larger vehicles, were the predominant conflict type during filtering, with sideswipe risk about three times that of non-filtering movements.

Three points can be taken from these studies with reasonable confidence. The speed differential between the motorcycle and surrounding traffic modifies injury severity when a splitting or filtering crash occurs, with risk rising above roughly 15 mph of differential; this is the evidence most directly relevant to the speed caps in the recent laws. Riders who split lanes in the California data also differ from other crash-involved riders in ways that independently predict better outcomes, so part of their favorable injury profile has nothing to do with the maneuver. And where filtering is common, same-direction sideswipe conflicts predominate. The evidence does not establish whether legalizing filtering changes the probability of crashing. The California study is a case-only design: it compares outcomes among riders who crashed, has no exposure denominator, and cannot be used to estimate crash risk per mile or per maneuver [35, 43]. The Ouellet [29] comparison of observed prevalence with crash-involved share is the closest approximation to an exposure-adjusted estimate in the U.S. literature, but it rests on 1970s crash data and a limited set of camera sites.

RQ2: Rider and driver behavior. A small set of naturalistic, instrumented, and trajectory-based studies describes how riders filter rather than how often they crash. Instrumented and eye-tracking studies in the Paris region and in Greece describe filtering as a routine but perceptually demanding activity in which riders continuously search for hazards and monitor the vehicles ahead, using surrounding vehicles not only to hold position but to anticipate drivers' movements [7, 31]. An on-road study in Melbourne found that riders who filtered on the approach to a red signal attended more to signal timing and their own maneuver and less to through and following traffic than riders who did not; filtering occurred in 27 percent of 43 eligible queue opportunities [27]. A study at a signalized intersection in Thailand found that the lateral clearance riders accept varies with motorcycle speed, filtering side, the type and movement state of the adjacent vehicle, and rider characteristics [32]. Filtering, on this evidence, is not a uniform maneuver: riders modify speed and positioning in response to surrounding vehicles, traffic conditions, and available space.

Among the seed studies, only one source measured filtering both by self-report and by direct observation in the same setting. In the Sydney trial, 63 percent of responding motorcyclists reported that they always or mostly filtered, while video observation at monitored sites identified filtering among about 20 to 30 percent of observed riders [38]. Because the samples and contexts differed, the gap is not direct evidence of self-report bias, but it shows why self-reported behavior must be triangulated with observation. The Western Australian review adds a compliance perspective: most respondents had observed riders not complying with the new rules, yet almost half of surveyed riders reported changing their behavior to comply, and offence data showed no notable change after legalization [42]. None of the seed studies asks the question that matters most for policy evaluation, which is whether rider or motorist behavior changes when the law changes.

RQ3: Awareness, knowledge, and attitudes. Beanland et al. [8] surveyed 249 drivers shortly before a two-year filtering trial began in the Australian Capital Territory. Although 61 percent reported observing filtering at least weekly, 28 percent incorrectly believed it was already legal, and 70 percent believed it should remain illegal. Those who believed filtering was already legal (odds ratio 4.67) and those who rode powered two-wheelers themselves (odds ratio 4.87) were more likely to support legalization. Support was associated with perceived congestion and safety benefits and the difficulty of enforcing a prohibition; opposition centered on safety concerns, including difficulty detecting motorcycles in blind spots. In the Sydney trial, 48 percent of surveyed motorcyclists reported feeling safer while filtering and about a quarter reported feeling less safe, and interactions with pedestrians crossing between queued vehicles were identified as a concern [38]. The Western Australian community knowledge test indicated that a large share of the public lacked accurate knowledge of the law even after legalization, although most riders were aware of the rules [42]. Research from the second French experiment shows that social acceptability varies with prior experience of the maneuver and with reported compliance [9]. Taken together, these studies indicate that frequent exposure to filtering does not ensure accurate knowledge of either the maneuver or its legal status, and that perceived legality and actual legal knowledge are easily conflated. All of this evidence comes from outside the United States and mostly from cross-sectional or convenience samples.

RQ4: Effects of legalization. Three government evaluations were in the seed set, and all three are agency reports rather than peer-reviewed studies; a fourth, from Belgium, was retrieved by the search and is not yet synthesized [23]. The 2013 Sydney trial ran for eight weeks at five treatment sites and one control site, combining video observation, traffic sensors, intercept and online surveys, and police monitoring; no crashes occurred during the trial, and its findings led to statewide legalization under a 30 km/h limit [38]. The Western Australian review, conducted 18 months or more after the March 2021 amendments, found no notable change in rear-end or sideswipe crashes involving motorcyclists and no notable change in offences, while documenting incomplete community knowledge of the rules [42]. The French experiment, covering 11 departments from February 2016 to January 2021 with a control zone and about 4,500 police reports, found that crashes involving powered two-wheelers rose by about 12 percent on the experimental networks while falling by about 10 percent across the experimental departments as a whole, that compliance with the 50 km/h cap improved from 23 to 40 percent, and that the overall picture was not sufficient to justify codifying the practice without further experimentation [11]. Cerema identified excessive motorcycle speed, unexpected lane changes, and failures by other road users to comply with traffic rules as recurring factors in crashes during inter-file riding. The three evaluations differ in duration, crash counts, and context, and none adjusts for exposure, so they do not support a pooled estimate; what they share is a design lesson, that legalization is a regulatory and a behavioral intervention at once, and an evaluation that counts only crashes will miss much of what changed. In the United States, Colorado's statute requires the state transportation agency to examine rear-end and sideswipe crashes before and after implementation [12], which creates a defined opportunity for the first such evaluation of a U.S. law.

3.4 Legal Census: Results for 51 Jurisdictions

Table 5 reports the coding of every state and the District of Columbia, with the governing provision, the basis of the code, the second official source, and the verification status; Figure 3 maps the result for each maneuver. All 51 jurisdictions were verified from the primary legal text on the official legislature, code, or agency site; 44 also have a second official source, and for seven no official second source could be located. The ten pilot jurisdictions were Alaska [4], Arkansas [6], California [10], the District of Columbia [13], Hawaii [16], Idaho [19], Illinois [20], Minnesota [24], Montana [26], and Utah [39, 40]; the coding scheme did not need to be changed after the pilot.

Table 5: Legal Census of Lane Filtering and Lane Splitting in the 50 States and the District of Columbia

Jurisdiction	Governing provision	Filtering	Splitting	Basis of the code	Second official source	Status
Alabama	Ala. Code § 32-5A-242(b)-(c)	C	C	Same-lane overtaking and operation between lanes or adjacent lines or rows barred; police exception	Official source agrees	Verified
Alaska	13 AAC 02.427(b)-(c)	C	C	Same-lane overtaking and operation between lanes or adjacent lines or rows barred; police exception	Official manual, general wording	Verified
Arizona	A.R.S. § 28-903(B), (C), (E), (F)	A	C	Filtering past a stopped vehicle in the same lane and between lanes at 15 mph or less, on a road with two or more same-direction lanes and a posted limit of 45 mph or less; otherwise same-lane passing and riding between lanes barred (SB 1273, 2022). A 2026 law makes violations chargeable as reckless driving (§ 28-693)	Official source agrees	Verified
Arkansas	Ark. Code § 27-51-302	D	D	General single-lane rule only; no motorcycle-specific provision	None located	Verified (primary only)
California	Cal. Veh. Code § 21658.1	B	B	Defines lane splitting (two-wheeled motorcycle between rows of stopped or moving vehicles) and authorizes CHP guidelines; no express authorization clause; AB 51 (2016)	Official source agrees	Verified
Colorado	C.R.S. § 42-4-1503(3)(a)-(b)	A	C	Two-wheeled motorcycle may pass in the same lane when adjacent same-direction traffic is stopped, lanes wide enough, 15 mph or less; between-lanes riding otherwise barred; SB 24-079; repealed 1 Sep 2027 after CDOT report	Official source agrees	Verified

Continued on next page

Table 5 (continued)

Jurisdiction	Governing provision	Filtering	Splitting	Basis of the code	Second official source	Status
Connecticut	Conn. Gen. Stat. § 14-289b(b)	C	C	Same-lane overtaking and operation between lanes or adjacent lines or rows barred; police exception	Official manual, general wording	Verified
Delaware	21 Del. C. §§ 4122, 4185	D	D	General single-lane rule only; no motorcycle-specific provision	Official manual, general wording	Verified
District of Columbia	18 DCMR §§ 2201.6(a), 2202, 2215	D	D	General single-lane and overtaking rules only; motorcycle section has no lane-position rule	Official manual, general wording	Verified
Florida	Fla. Stat. § 316.209(2)-(3)	C	C	Same-lane overtaking and operation between lanes or adjacent lines or rows barred; police exception	Official source agrees	Verified
Georgia	O.C.G.A. § 40-6-312(b)-(c)	C	C	Same-lane overtaking and operation between lanes or adjacent lines or rows barred; police exception	Official manual, general wording	Verified
Hawaii	Haw. Rev. Stat. § 291C-153(b)	C	C	Same-lane overtaking and operation between lanes or adjacent rows barred; a 2019 designated-shoulder provision sunset 31 Dec 2020	Official source agrees	Verified
Idaho	ITD Motorcycle Rider's Manual, p. 44	E	E	No motorcycle-specific provision; ITD rider manual and State Police state both maneuvers are not legal	Official source agrees	Verified
Illinois	625 ILCS 5/11-703(c); 5/11-704(b)	C	C	Two-wheeled vehicle may not pass between two same-direction vehicles unless an unobstructed lane is available; may not pass on the right without 8 ft of pavement	Identified, not reviewed	Verified
Indiana	Ind. Code § 9-21-10-6	D	D	General single-lane rule only; no motorcycle-specific provision	Official manual, general wording	Verified
Iowa	Iowa Code § 321.275	C	C	Same-lane overtaking and operation between lanes or adjacent lines or rows barred; police exception	Official manual, general wording	Verified
Kansas	K.S.A. 8-1595(b)-(c)	C	C	Same-lane overtaking and operation between lanes or adjacent lines or rows barred; police exception	Official source agrees	Verified
Kentucky	KRS 189.285; 189.340	D	D	General single-lane rule only; no motorcycle-specific provision	Official manual, general wording	Verified
Louisiana	La. R.S. 32:191.1(B)-(C)	C	C	Same-lane overtaking and operation between lanes or adjacent lines or rows barred; police exception	Official manual, general wording	Verified
Maine	29-A M.R.S. § 2062(5)-(6)	C	C	Same-lane overtaking and operation between lanes or adjacent lines or rows barred; police exception	None located	Verified (primary only)
Maryland	Md. Code, Transp. § 21-1303(c)-(d)	C	C	Same-lane overtaking and operation between lanes or adjacent lines or rows barred; police exception	Official source agrees	Verified
Massachusetts	Mass. Gen. Laws ch. 89, § 4A	C	C	Shall not pass any other motor vehicle within the same lane except a motorcycle; single-lane rule (C via same-lane-passing provision)	Official source agrees	Verified
Michigan	MCL 257.660(5)	C	C	Shall not pass between lines of traffic; may pass in an unoccupied lane	Official source agrees	Verified
Minnesota	Minn. Stat. § 169.974 subd. 5(f)-(h)	A	A	Same-lane passing and riding between lanes permitted at 25 mph or less and no more than 15 mph above traffic (so 15 mph when traffic is stopped); excludes roundabouts, school zones, single-lane work zones, on-ramps with queued traffic; effective 1 Jul 2025	Official source agrees	Verified
Mississippi	Miss. Code Ann. § 63-3-603(a)	D	D	General single-lane rule only; no motorcycle-specific provision	Official manual, general wording	Verified
Missouri	Mo. Rev. Stat. § 304.015	D	D	General single-lane rule only; no motorcycle-specific provision	Official manual, general wording	Verified

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Table 5 (continued)

Jurisdiction	Governing provision	Filtering	Splitting	Basis of the code	Second official source	Status
Montana	Mont. Code Ann. § 61-8-392; §§ 61-8-328(1), 61-8-359(6)-(7)	A	D	Statute defines lane filtering: passing a vehicle stopped or moving 10 mph or less, in the same lane, motorcycle 20 mph or less, lanes wide enough (SB 9, 2021). No provision reaches traffic above 10 mph, so splitting is D	None located	Verified (primary only)
Nebraska	Neb. Rev. Stat. § 60-6,308(2)-(3)	C	C	Same-lane overtaking and operation between lanes or adjacent lines or rows barred; police exception	Official source agrees	Verified
Nevada	NRS 486.351(1)-(2)	C	C	No driving between moving or stationary vehicles in adjacent lanes; no passing within the same lane	Official source agrees	Verified
New Hampshire	N.H. RSA 265:121 II-III, V	C	C	Same-lane overtaking and operation between lanes or adjacent lines or rows barred; police exception	Identified, not reviewed	Verified
New Jersey	N.J.S.A. 39:4-88(b)	D	D	General single-lane rule only; no motorcycle-specific provision	Official manual, general wording	Verified
New Mexico	NMSA 1978 § 66-7-317	D	D	General single-lane rule only; no motorcycle-specific provision	Official law compilation (2009-10), general wording	Verified
New York	N.Y. Veh. & Traf. Law § 1252(b)-(c)	C	C	Same-lane overtaking and operation between lanes or adjacent lines or rows barred; police exception	Official source agrees	Verified
North Carolina	N.C. Gen. Stat. § 20-146.1(a)-(b)	D	D	General single-lane rule only; no motorcycle-specific provision	Official manual, general wording	Verified
North Dakota	N.D.C.C. § 39-10.2-03(2)-(3)	C	C	Same-lane overtaking and operation between lanes or adjacent lines or rows barred; police exception	Official manual, general wording	Verified
Ohio	Ohio Rev. Code §§ 4511.33, 4511.55(B)	D	D	General single-lane rule only; no motorcycle-specific provision	Official source agrees	Verified
Oklahoma	47 O.S. § 11-1103(D)	C	C	Shall not pass other vehicles between lanes of traffic traveling in the same direction	None located	Verified (primary only)
Oregon	ORS 814.240(1)(a)-(b)	C	C	Same-lane overtaking and operation between lanes or adjacent lines or rows barred; police exception	Official source agrees	Verified
Pennsylvania	75 Pa.C.S. § 3523(b)-(c)	C	C	Same-lane overtaking and operation between lanes or adjacent lines or rows barred; police exception	Official manual, general wording	Verified
Rhode Island	R.I. Gen. Laws § 31-15-11(1)	D	D	General single-lane rule only; no motorcycle-specific provision	None located	Verified (primary only)
South Carolina	S.C. Code § 56-5-3640(b)-(c)	C	C	Same-lane overtaking and operation between lanes or adjacent lines or rows barred; police exception	Official source agrees	Verified
South Dakota	S.D. Codified Laws § 32-20-9.3	C	C	No operation between adjacent lanes or adjacent lines or rows of vehicles	Identified, not reviewed	Verified
Tennessee	Tenn. Code Ann. § 55-8-182(b)-(c)	C	C	Same-lane overtaking and operation between lanes or adjacent lines or rows barred; police exception	Official source agrees	Verified
Texas	Tex. Transp. Code § 545.0605(a)(3)(B)-(C)	C	C	May not operate between lanes of same-direction traffic or pass a vehicle in the same lane (HB 4122, 2023)	None located	Verified (primary only)
Utah	Utah Code §§ 41-6a-102(39)-(40), 41-6a-704(6), 41-6a-704.1, 41-6a-1502(2)-(3)	A	C	Filtering past stopped traffic at 15 mph or less on a road with two or more same-direction lanes and posted limit 45 mph or less (off-ramp exception, on-ramps excluded); sunset 1 Jul 2027. Lane splitting defined and expressly prohibited (HB 190, 2025)	Official source agrees	Verified
Vermont	23 V.S.A. § 1115(b)-(c)	C	C	Same-lane overtaking and operation between lanes or adjacent lines or rows barred; police exception	Official manual, general wording	Verified

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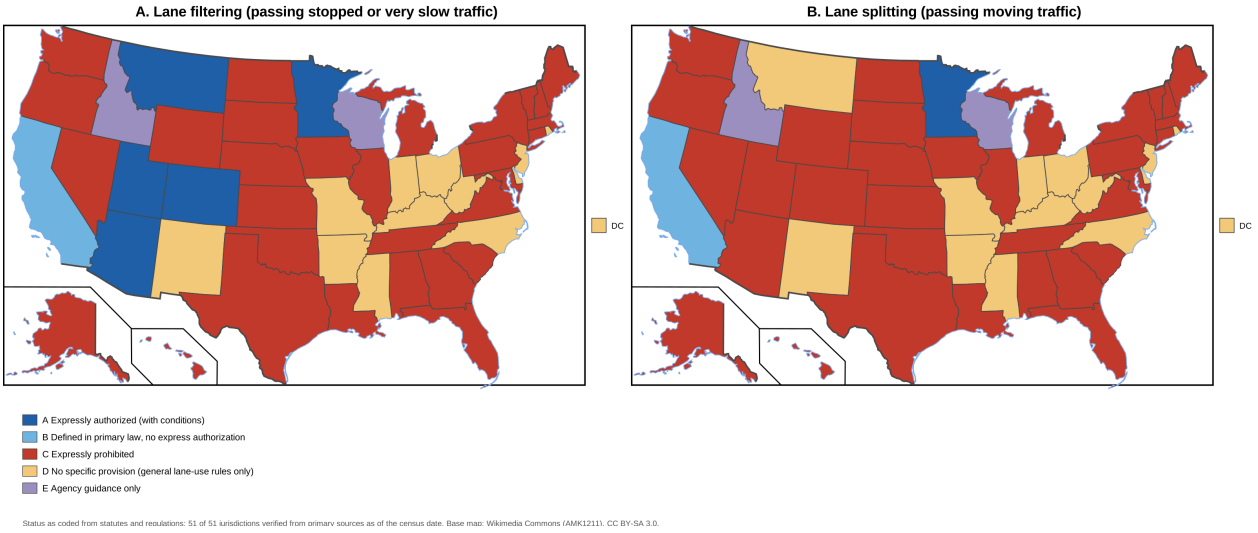


Figure 3: Legal status of (A) lane filtering and (B) lane splitting in the 50 states and the District of Columbia, coded from the primary legal text (Table 5; census dates 26 September to 7 October 2026). Base map: Wikimedia Commons (AMK1211), CC BY-SA 3.0.

Table 5 (continued)

Jurisdiction	Governing provision	Filtering	Splitting	Basis of the code	Second official source	Status
Virginia	Va. Code § 46.2-857	C	C	Reckless driving to drive abreast of another vehicle in a lane designed for one vehicle; no between-lanes wording (C via same-lane-passing provision; Virginia DMV states it prohibits filtering)	Official source agrees	Verified
Washington	RCW 46.61.608(2)-(3)	C	C	Same-lane overtaking and operation between lanes or adjacent lines or rows barred; police exception	Official source agrees	Verified
West Virginia	W. Va. Code § 17C-7-9(a)(1)	D	D	General single-lane rule only; no motorcycle-specific provision	Official manual, general wording	Verified
Wisconsin	WisDOT motorcycle guidance; Wis. Stat. § 346.595(1)	E	E	No motorcycle-specific provision (§ 346.595 is a full-lane-use rule); WisDOT states white lining or lane splitting is illegal	None located	Verified (primary only)
Wyoming	Wyo. Stat. § 31-5-115(f)-(g)	C	C	Same-lane overtaking and operation between lanes or adjacent lines or rows barred; police exception	Identified, not reviewed	Verified

Note: A = expressly authorized; B = defined in primary law without an express authorization provision; C = expressly prohibited; D = no specific provision identified, general lane-use rules only; E = official agency guidance only. Category D records the absence of a provision and is not a prohibition. “Verified (primary only)” means the primary text was read on the official site but no separate official second source was found. “Official manual, general wording” means the state motorcycle operator manual discourages lane sharing in generic terms without citing the state’s provision. Primary text read 26 September to 7 October 2026 on official legislature, code, or agency sites; citations give the operative subsections and omit the police-exception subsection; full citations, URLs, access dates, and notes are in the census workbook. “Official source agrees” includes two non-legislative official sources (an FDOT-funded program for Florida and a municipal police department for Arizona).

For lane filtering, five states expressly authorize the maneuver with conditions (category A: Arizona, Colorado, Minnesota, Montana, Utah), one defines it in primary law without an express authorization clause (B: California), 30 expressly prohibit it (C), 13 have no motorcycle-specific provision and only general lane-use rules (D), and two have agency guidance stating that it is illegal without a governing provision (E: Idaho, Wisconsin). For lane splitting, one state authorizes it (A: Minnesota), one defines it (B: California), 33 expressly prohibit it (C), 14 have no specific provision (D), and two have agency guidance only (E). The two maneuvers are coded identically in 47 jurisdictions; the four exceptions are filtering states, where Utah, Arizona, and Colorado expressly prohibit splitting while authorizing filtering and Montana’s statute reaches only traffic stopped or moving at 10 mph or less, so that splitting falls to category D. Minnesota, the fifth filtering state, authorizes both maneuvers under one rule and is coded A for both. Six jurisdictions therefore have at least one form of riding between lanes that is authorized or defined in primary law, and 45 have none.

The category D group deserves comment because it is the one that secondary summaries most often describe as “illegal”. In Arkansas, Delaware, the District of Columbia, Indiana, Kentucky, Mississippi, Missouri, New

Jersey, New Mexico, North Carolina, Ohio, Rhode Island, and West Virginia, and for splitting also Montana, the only applicable provision is a general rule that a vehicle be driven as nearly as practicable within a single lane, or a motorcycle full-lane-use and two-abreast rule; none of these names riding between lanes or passing within a lane. Whether such a rule is enforced against a filtering motorcyclist is a question of enforcement practice and case law that the census does not reach. The category E jurisdictions are the mirror image: Idaho and Wisconsin have the same kind of general rule, but their transportation agencies state publicly that lane filtering and lane splitting are illegal, so the agency position is recorded as the governing source. The category C group is more uniform than its size suggests: 24 of the 30 filtering prohibitions use the Uniform Vehicle Code pair of provisions, one barring overtaking in the same lane as the overtaken vehicle and one barring operation between lanes of traffic or between adjacent lines or rows of vehicles, usually with a police exception. The exceptions are Illinois, whose provision bars a two-wheeled vehicle from passing between two vehicles unless an unobstructed lane is available; Michigan, Oklahoma, and South Dakota, which bar riding or passing between lanes or lines of traffic without a separate same-lane clause; and Massachusetts and Virginia, which bar passing or driving abreast within a single lane without naming riding between lanes. Texas is the most recent addition to the group: its express prohibition was enacted in 2023 (HB 4122, effective 1 September 2023), after the first wave of authorizing statutes elsewhere.

Table 6 compares the authorizing provisions. The five statutes agree on little beyond the idea of a motorcycle speed cap. Utah, Arizona, and Colorado cap the motorcycle at 15 mph and require the overtaken traffic to be stopped (in Arizona the between-lanes clause is not expressly tied to stopped traffic in the text, although the same-lane clause is); Montana caps the motorcycle at 20 mph and allows the overtaken traffic to be moving at 10 mph or less; Minnesota sets a single rule of 25 mph or less and no more than 15 mph faster than traffic, which caps filtering past stopped traffic at 15 mph and permits passing moving traffic up to 25 mph. Utah and Arizona require at least two same-direction lanes and a posted limit of 45 mph or less; Colorado requires that adjacent traffic be stopped and the lanes wide enough; Minnesota excludes roundabouts, school zones, single-lane work zones, and freeway on-ramps with queued traffic; Utah excludes on-ramps and extends the provision to off-ramps. Utah's provision sunsets on 1 July 2027 and Colorado's is repealed on 1 September 2027 after a required CDOT report on rear-end and sideswipe crashes. Arizona added, in 2026, that a violation of the filtering conditions is chargeable as reckless driving, without changing the conditions themselves. California's provision, which predates all of these, defines lane splitting between rows of stopped or moving vehicles and sets no speed cap. An evaluation that treated these six jurisdictions as one "permitted" group would be pooling at least four different treatments.

Table 6: Conditions Attached to the Authorizing and Defining Provisions

Jurisdiction	Maneuver	Motorcycle speed cap	Overtaken traffic	Roadway conditions	Excluded locations	Effective	Sunset or review
Utah	Filtering (A); splitting expressly prohibited (C)	15 mph	Stopped	Two or more same-direction lanes; posted limit 45 mph or less (not required on off-ramps)	On-ramps	14 May 2019 (current version 1 May 2024)	Sunset 1 Jul 2027
Montana	Filtering (A); no provision for moving traffic (D)	20 mph	Stopped or 10 mph or less	Lanes wide enough to pass safely; same lane; two-wheeled only	Not stated	1 Oct 2021	None
Arizona	Filtering (A); splitting expressly prohibited (C)	15 mph	Stopped	Two or more same-direction lanes; posted limit 45 mph or less; two-wheeled only	Not stated	24 Sep 2022	None; 2026 law makes violations chargeable as reckless driving
Colorado	Filtering (A); splitting expressly prohibited (C)	15 mph	Adjacent same-direction traffic stopped	Lanes wide enough; two-wheeled only	Right shoulder; right of the farthest-right lane; opposing-direction lane	7 Aug 2024	Repealed 1 Sep 2027 after CDOT crash report
Minnesota	Filtering and splitting (A/A), one rule	25 mph, and no more than 15 mph faster than traffic (15 mph when traffic is stopped)	Stopped or moving	Not stated	Roundabouts; school zones; single-lane work zones; freeway or expressway on-ramps with queued traffic	1 Jul 2025	None
California	Lane splitting defined (B/B); no express authorization clause	None in statute (CHP guidelines)	Stopped or moving	Two-wheeled only	Not stated	1 Jan 2017	None

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Table 6 (continued)

Jurisdiction	Maneuver	Motorcycle speed cap	Overtaken traffic	Roadway conditions	Excluded locations	Effective	Sunset or review

Note: Conditions are summarized from the statutory text as coded in the census workbook; the statutes also contain general safety language (for example, that the movement be made safely) that is not listed here.

The conflict log recorded nine items (Table 7), and the notes recorded a further set of discrepancies that did not involve the starting dataset. Two conflict items corrected secondary sources: a commercial blog places the end of Hawaii's designated-shoulder provision in 2019, whereas the session law shows it took effect on 1 January 2019, sunset on 31 December 2020, and never authorized riding between lanes; and the starting dataset grouped Montana with states that permit passing moving traffic, whereas the statute defines the authorized act as lane filtering past stopped or very slow vehicles and no other provision reaches moving traffic. Two items were differences of category label rather than substance (California, Illinois), one concerned the location of the governing provision (Utah), one was an agency-wording discrepancy that the statute resolved (Minnesota), one was a gap in the starting dataset (the District of Columbia), one was a category decision confirmed against the Idaho precedent (Wisconsin, D to E), and one was a 2026 amendment that turned out to affect only the penalty section (Arizona). In every jurisdiction that the starting dataset covered, its substantive conclusion about whether the maneuver is permitted was confirmed by the primary text, with Montana as the one case where its description of the maneuver was not supported; the discrepancies concerned category boundaries, the location of the governing provision, and dates. The compiled table was a good starting point but not a substitute for the statute.

The coder's notes also document errors in secondary sources that are not conflicts with the starting dataset but that illustrate why the census was needed. A law-firm summary states that Delaware expressly bans lane splitting; the Delaware Code motorcycle section covers seats, passengers, helmets, and handgrips and contains no lane rule. A similar summary paraphrases a Kentucky lane-splitting ban in KRS 189.285(3); the official text of that section contains no such provision. Another states that Missouri's § 304.015 expressly bars lane splitting; the official text is the general single-lane rule. A summary describes lane splitting as now legal in Mississippi; the 2016 bill that would have created a stalled-traffic exception was not enacted. A summary reports that Texas enacted limited filtering in 2025; the bill died in committee and the 2023 prohibition stands. A rider website quotes a Rhode Island between-lanes ban that is, word for word, the New Hampshire statute. Several summaries cite a municipal ordinance as New Mexico's state-law ban, and give conflicting state citations. Together with the Hawaii date and the Montana grouping, the census documents errors in secondary summaries for nine jurisdictions, in each case in the direction of describing the law as more specific, more permissive, or more recent than it is.

The notes also record legislative activity that a static map does not show. Bills to authorize filtering or splitting were introduced and not enacted since 2021 in at least ten states: Connecticut (SB 1083, 2025), Missouri (SB 1369, 2026), New Jersey (A4668, 2022), Ohio (HB 776, pending in committee), Oklahoma (HB 2667, 2022), Oregon (SB 574, vetoed in 2021; SB 422, 2023; HB 3542, 2025), Rhode Island (S 2209, 2024; H 5658, 2025; H 7354, 2026), Tennessee (SB 298 and HB 1454, 2023), Texas (HB 2957, 2025), and Virginia (SB 435, 2026). Two bills ran the other way: Arizona's SB 1412 (2023) would have repealed the 2022 law and was not enacted, and West Virginia's HB 4972 (2026) would add an express prohibition to a state that currently has none. Montana's own authorization followed an unenacted 2017 bill. The legal landscape is therefore not static, and the census dates (26 September to 7 October 2026, with 39 of the 51 jurisdictions read on 7 October) are part of the result.

Table 7: Conflict Log for the Legal Census

ID	Jurisdiction	Item	Sources compared	Resolution against the primary text
CL-001	Montana	Is the authorized maneuver filtering or splitting?	Starting dataset groups Montana with states allowing passing of moving traffic; statute defines "lane filtering"	Filtering = A under the statute's own term; no provision reaches traffic above 10 mph, so splitting = D
CL-002	Hawaii	Year the designated-shoulder authority expired	Starting dataset: 2020; commercial blog: 2019	Act 218 took effect 1 Jan 2019 and sunset 31 Dec 2020; the 2019 claim is incorrect; the provision never authorized riding between lanes
CL-003	Minnesota	Separate speed cap for filtering versus splitting?	Starting dataset: single rule; agency page: 15 mph filtering, 25 mph splitting	Statute sets one rule (25 mph or less and no more than 15 mph over traffic); with stopped traffic the differential caps filtering at 15 mph; no conflict in substance

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Table 7 (continued)

ID	Jurisdiction	Item	Sources compared	Resolution against the primary text
CL-004	California	Classification of § 21658.1	Starting dataset: both "Permitted"	Statute defines lane splitting across stopped and moving traffic but has no express authorization clause; coded B/B
CL-005	Illinois	Whether § 11-703(c) is a motorcycle-specific restriction	Starting dataset: "Not permitted"	Provision is specific to two-wheeled vehicles and expressly bars passing between two vehicles; coded C/C; difference is category label only
CL-006	Utah	Location of the lane-splitting prohibition	Starting dataset cites § 41-6a-704 only	Prohibition sits in §§ 41-6a-102(40), 41-6a-704.1, and 41-6a-1502; coding (A/C) confirmed; additional sections cited
CL-007	District of Columbia	Jurisdiction not covered by the starting dataset	Starting dataset covers 50 states only	Coded D/D from 18 DCMR; DMV manual discourages lane sharing in general terms but cites no DC rule, so category E does not apply
CL-008	Wisconsin	Category label: no specific provision versus agency guidance (D or E)	Starting dataset: "No express permission"; WisDOT states white lining or lane splitting is illegal	Coded E/E, consistent with Idaho: agency position without a specific governing provision; not a substantive disagreement
CL-009	Arizona	2026 amendment reported by a municipal police department	Starting dataset: 2022 law only; police page: violations chargeable as reckless driving from 12 Sep 2026	Laws 2026, ch. 261 (HB 2941) amends § 28-693 (reckless driving) only; § 28-903 and its filtering conditions are unchanged; coding and dates stand

Note: Secondary sources appear in this table only as objects of comparison; no legal status in the census is taken from a secondary source.

4 Discussion

Two kinds of result are available at this stage, and they should be kept apart. The first is methodological: what the search and the census revealed about how the evidence on this topic is organized. The second is substantive: what the seed studies, read together, suggest about the safety of filtering and about what an evaluation of the U.S. laws would need to measure. The substantive findings are preliminary and will be revised when the systematic search is complete.

The methodological results concern retrieval. The terminology of this literature is unstable: the same maneuver is called lane filtering, lane splitting, lane sharing, white-lining, inter-file riding, seepage, and simply "filtering", and the statutes add their own definitions. A search built on "lane filtering" and "lane splitting" missed the most direct quantitative conflict study in the field, and the revised search that captures it also retrieves exhaust filters, Kalman filters, and radiation dosimeters. The practical lesson, which the search log documents, is that for this topic a sensitive search with noise removed at screening is preferable to a precise search that loses relevant records, and that validation against known studies should precede screening rather than follow it. A second methodological result is that the three agency evaluations of legalization that the authors knew of before the search, which are the main before-and-after evidence in the seed set, were not retrieved by any of the seven searches and do not appear in the record set; the search did retrieve a Belgian evaluation that was published as a conference paper [23], which suggests that evaluations reach the indexes only when they pass through conference or journal channels, and that a review relying on database searching alone would miss most of them. A third is that TRID, which is often omitted from systematic reviews in favor of the two large multidisciplinary indexes, contributed 39 of the 114 unique records, including the California crash study that anchors the field.

The substantive picture from the seed studies is consistent across sources, but thin. The speed differential between the motorcycle and surrounding traffic is the best-supported modifier of injury severity, and it is the parameter that every recent U.S. statute regulates, either directly (Minnesota's 15-mph differential) or through a motorcycle speed cap combined with a requirement that traffic be stopped or nearly so. The favorable injury profile of lane-splitting riders in California is real but partly explained by who those riders are rather than by what they are doing. Filtering is a skilled, attention-demanding maneuver whose characteristic conflict is the same-direction sideswipe, which is low-severity, often property-damage-only, and therefore under-reported. Road users' knowledge of the law is poor even where the maneuver is common, and perceived legality predicts support for legalization. And the three evaluations of legalization in the seed set report mixed or null crash effects but substantial change in behavior and awareness, within short observation periods and without exposure adjustment. None of this establishes whether legalization changes crash risk. The rear-end protection that motivates the laws and the sideswipe exposure that worries their critics are both plausible, and because they can offset one another, an aggregate crash count could be null while both effects are real.

The design implication for an evaluation of the U.S. laws, which is a recommendation of this review rather than a finding of the literature, is that legalization should not be treated as a single intervention. The census shows that the statutes differ in the permitted motorcycle speed, in whether surrounding traffic must be stopped or may be moving slowly, in the roadways where filtering is allowed, and in whether splitting is prohibited, defined, or permitted alongside it. An evaluation that codes states only as permitting or prohibiting filtering could attribute to filtering in general effects that belong to specific conditions. The census also shows why the legal variable has to come from the primary text: the categories that secondary sources use ("legal", "illegal", "not permitted") do not distinguish an express prohibition (30 jurisdictions) from the absence of a provision (13) or from agency guidance alone (2), and the census found errors of date, citation, or substance in circulating summaries for nine jurisdictions. For an evaluation, the 13 category D jurisdictions are the awkward group: they cannot be coded as prohibiting filtering on the basis of the statute, yet enforcement practice may treat them as if they did, which is a reason to measure filtering prevalence directly rather than to infer it from the law. A controlled before-and-after or interrupted time-series design with comparison states, the approach used by the French and Western Australian evaluations, is the most suitable option, with exposure treated as a first-order measurement problem rather than a nuisance covariate, since motorcycle vehicle-miles are poorly estimated and riding is strongly seasonal in several of the states that have legalized filtering.

Four gaps, identified from the seed studies and to be tested against the full record set, organize the research agenda. None of the studies known to the authors follows crash, travel, and behavioral outcomes through a U.S. lane-filtering law change. The prevalence of filtering before any of these laws took effect is unmeasured, which matters because a statute may formalize existing behavior instead of creating new exposure. Little evidence links what road users know about the law to how they behave, and almost none concerns the drivers who share the road with filtering motorcycles. And no study known to the authors compares statutes with different speed and traffic-condition thresholds to ask whether the restrictions themselves make a difference.

5 Limitations and Future Work

This is a stage-one report, and its limitations follow from that. The systematic search has been run and validated, but screening, full-text assessment, and extraction have not, so the record set in Figure 2 is a set of candidates, not of included studies, and the synthesis in the Results is based on the seed set rather than on the systematic search. The seed set was assembled from prior reviews and the authors' knowledge of the field and is likely to over-represent well-known studies and under-represent recent work from Asia and Europe; the search results already indicate that the latter is substantial. The other-methods track has not been run, so the agency evaluations are represented only by the three the authors knew of. The legal census is complete, but its date fields are not: an original effective date could be confirmed from the official history line for only 10 jurisdictions and a current-version date for 16, because most official histories give a session law or year rather than a date, and the census recorded nothing it could not confirm. In 18 jurisdictions the second official source agrees only partially with the primary text (17 motorcycle operator manuals and one dated agency compilation of New Mexico law), because these documents use generic safety wording rather than the state's own rule. The coding of Massachusetts and Virginia as category C rests on same-lane-passing provisions that do not name riding between lanes, and the coding of Idaho and Wisconsin as category E rests on agency statements rather than a governing provision; both choices are documented so that a reader who prefers a different rule can recode them. The dedup procedure matched identifiers and exact titles and will be repeated in a reference manager before screening, so the count of 114 may change by a small number. Finally, the review is limited to English-language records and to statutory and regulatory text, and cannot see case law, local ordinances, or enforcement practice.

The next steps are fixed by the protocol. Two reviewers will screen the 114 records after calibration on 50, the other-methods searches will be run on the dates recorded in the search log, full texts will be assessed with coded exclusion reasons, and the extraction form will be completed for every included study. The census will be re-checked against the official sites immediately before the updated version is posted, because at least two of the filtering statutes carry 2027 sunset dates and bills are pending in several states. The updated version of this paper will report the completed PRISMA flow diagram with counts at every stage, the descriptive analysis of included studies, and the thematic synthesis by research question. Beyond the review, the evaluation of the U.S. laws that it is intended to support will require crash data with a lane-position identifier, exposure measures, direct observation of filtering prevalence before and after legalization, and repeated surveys of riders and drivers, combined within one design; none of the studies known to the authors has all four.

6 Conclusion

Five U.S. states have enacted laws expressly permitting motorcycle lane filtering since 2019, each under different speed and traffic-condition limits, and two of those provisions sunset in 2027. This paper puts on record the protocol for a systematic review of the evidence on lane filtering and lane splitting and a primary-source census of state law, reports a validated database search that identified 114 unique candidate records, presents preliminary findings from the seed studies, and reports the completed census of all 51 jurisdictions. The preliminary findings are that the speed differential is the clearest modifier of injury severity, that filtering is a demanding maneuver whose characteristic conflict is the same-direction sideswipe, that public knowledge of the law is poor, that legalization has not been shown to change crash frequency in the evaluations identified so far, and that the legal status of the maneuver is misdescribed often enough in secondary sources that it must be coded from the statutes. The census finds five authorizing states with at least four different sets of conditions, 30 express prohibitions, 13 jurisdictions with no motorcycle-specific provision, and two with agency guidance only. The completed review will test the preliminary findings against the systematic record set.

Author Contributions

The authors confirm their contributions to the paper as follows: study conception and design: Subasish Das, Iman Jamshidi; review protocol and search strategy: Iman Jamshidi, Tausif Islam Chowdhury; database searching, search validation, and search log: Tausif Islam Chowdhury; record compilation and duplicate removal: Iman Jamshidi; legal census and conflict log: Samia Sahid, with the pilot worked example and official-site confirmation of 13 jurisdictions by Iman Jamshidi; analysis and interpretation of results: Subasish Das, Iman Jamshidi, Tausif Islam Chowdhury, Samia Sahid; draft manuscript preparation: Iman Jamshidi; supervision: Subasish Das. All authors reviewed the results and approved the final version of the manuscript.

Declaration of Conflicting Interests

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Data Accessibility Statement

The search log, validation set, deduplicated record set, screening and extraction forms, and the legal census workbook with its conflict log are maintained as project workbooks and are available from the corresponding author on request; they will be deposited with the updated version of this paper. The full search strings are given in Appendix A.

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A Full Search Strings

The strings below are reproduced verbatim from the search log, as run. English-language limit applied in every database; no year or document-type limit.

SL-001. Scopus, v0.1, 01 October 2026, 19 records.

TITLE-ABS-KEY (("lane filtering" OR "lane-filtering" OR "lane splitting" OR "lane-splitting" OR "lane sharing" OR "filtering through traffic" OR "inter-file" OR "interfile" OR "white lining" OR "white-lining") AND (motorcycl* OR motorbike* OR "powered two-wheeler*" OR "powered two wheeler*" OR PTW* OR "two-wheeler*" OR moped* OR scooter*))

SL-002. Web of Science, v0.1, 01 October 2026, 22 records.

TS=(("lane filtering" OR "lane-filtering" OR "lane splitting" OR "lane-splitting" OR "lane sharing" OR "filtering through traffic" OR "inter-file" OR "interfile" OR "white lining" OR "white-lining") AND (motorcycl* OR motorbike* OR "powered two-wheeler*" OR "powered two wheeler*" OR PTW* OR "two-wheeler*" OR moped* OR scooter*))

SL-003. TRID, v0.1, 02 October 2026, 22 records.

"lane splitting"

SL-004. TRID, v0.1, 02 October 2026, 6 records.

"lane filtering"

SL-005. TRID, v0.1, 02 October 2026, 4 records.

"lane sharing" motorcycle

SL-006. TRID, v0.1, 02 October 2026, 34 records.

filtering motorcycle

SL-007. Scopus, v0.2, 06 October 2026, 41 records.

TITLE-ABS-KEY (("lane filtering" OR "lane-filtering" OR "lane splitting" OR "lane-splitting" OR "lane sharing" OR "filtering through traffic" OR "inter-file" OR "interfile" OR "white lining" OR "white-lining" OR "filtering maneuver*" OR "filtering manoeuvre*" OR "filtering behavio*" OR seepage OR (filtering W/3 (motorcycl* OR "two-wheeler*" OR PTW*))) AND (motorcycl* OR motorbike* OR "powered two-wheeler*" OR "powered two wheeler*" OR PTW* OR "two-wheeler*" OR moped* OR scooter*))

SL-008. Web of Science, v0.2, 06 October 2026, 68 records.

TS=(("lane filtering" OR "lane-filtering" OR "lane splitting" OR "lane-splitting" OR "lane sharing" OR "filtering through traffic" OR "inter-file" OR "interfile" OR "white lining" OR "white-lining" OR "filtering maneuver*" OR "filtering manoeuvre*" OR "filtering behavio*" OR seepage OR (filtering NEAR/3 (motorcycl* OR "two-wheeler*" OR PTW*))) AND (motorcycl* OR motorbike* OR "powered two-wheeler*" OR "powered two wheeler*" OR PTW* OR "two-wheeler*" OR moped* OR scooter*))

SL-009. TRID, v0.2, 06 October 2026, 22 records.

"lane splitting"

SL-010. TRID, v0.2, 06 October 2026, 6 records.

"lane filtering"

SL-011. TRID, v0.2, 06 October 2026, 4 records.

"lane sharing" motorcycle

SL-012. TRID, v0.2, 06 October 2026, 36 records.

filtering motorcycle

SL-013. TRID, v0.2, 06 October 2026, 1 records.

seepage motorcycle OR seepage "two-wheeler"

Draft 1, 10/12/2026